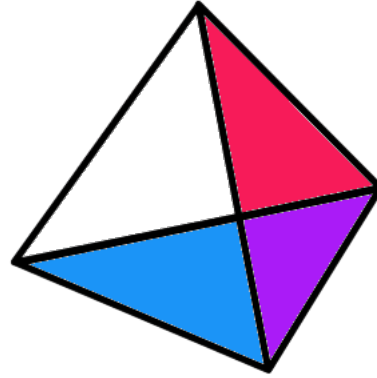




Project
PRISM

Genetic Variation

Genetics and Evolution

Genetic Variation

- Individuals within all species often vary in their phenotypic traits.
- Such phenotypic variations usually reflect **genetic variation**, differences among individuals in the composition of their genes or other DNA sequences.
- Some heritable phenotypic differences occur on an “either-or” basis (Mendel’s pea plants), such differences are typically determined by a single gene **locus***, with different alleles producing distinct phenotypes
- Other phenotypic differences vary in gradations along a continuum, usually resulting from the influence of two or more genes on a **phenotypic character****.
- Genetic variation may occur at the level of:
 - genes** (different alleles of the same locus),
 - whole chromosomes**,
 - or the **DNA nucleotide sequence** itself.
- Genetic variation is important because it provides the **raw material for evolution by natural selection**.

Genetic Variation vs Nonheritable Variation

- Not all phenotypic variation is genetic.
- Some differences between individuals are caused by **environmental factors** rather than differences in genotype.
- Therefore, only **heritable variation** contributes directly to evolutionary change.
- A population may look variable, but only the genetic component of that variation can be passed to offspring.

▼ **Figure 23.5 Nonheritable variation.** These caterpillars of the moth *Nemoria arizonaria* owe their different appearances to chemicals in their diets, not to differences in their genotypes. **(a)** Caterpillars raised on a diet of oak flowers resemble the flowers, whereas **(b)** their siblings raised on oak leaves resemble oak twigs.



Gene Variability and Nucleotide Variability

- Genetic variation occurs at two major levels:
 - gene variability** - variation in alleles at gene loci
 - nucleotide variability** - variation in the DNA base sequence
- Gene variability can be estimated by the **average percentage of loci that are heterozygous**.
- Nucleotide variability measures how different DNA sequences are between individuals.
- Much molecular variation does **not necessarily affect phenotype**.

How is Genetic Variability quantified?

Genetic variation at the whole-gene level (gene variability) can be quantified as the average percentage of loci that are heterozygous. (Recall that a heterozygous individual has two different alleles for a given locus, whereas a homozygous individual has two identical alleles for that locus.)

$$H_0 = \frac{\text{Number of Heterozygotes}}{\text{Total Number of Individuals}}$$

- A higher heterozygosity generally indicates greater genetic variation in a population.

(Do note that this formula is only used for Observed Heterozygosity, a more accurate way to determine genetic variability would be through allele frequency calculations)

How does Genetic Variation arise?

- In sexual reproduction, Three mechanisms contribute to the genetic variation arising from sexual reproduction:
 independent assortment* of chromosomes
 crossing over**
 and random fertilization***
- These three mechanisms give chromosomes their unique properties during meiosis and fertilization. Causing genetic variation between generations and individuals.

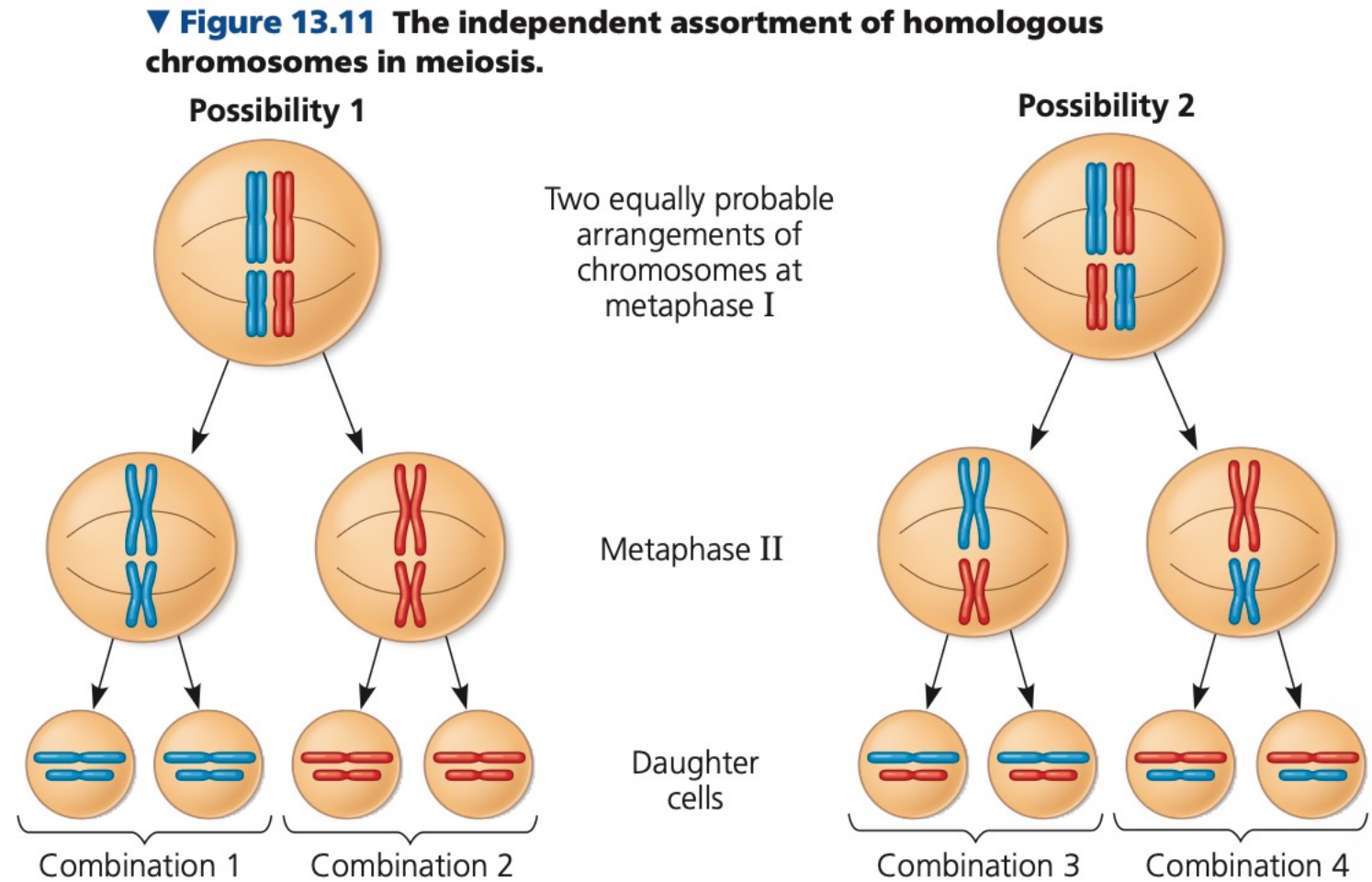
Genetic Variation at the Molecular Level

- DNA sequences can vary greatly even when phenotype changes little or not at all.
- A base-pair substitution may have no effect if:
 - it occurs in a **noncoding region**, or
 - it changes a codon without changing the amino acid because of the **redundancy of the genetic code**.
- Thus, molecular variation is often more extensive than visible phenotypic variation.

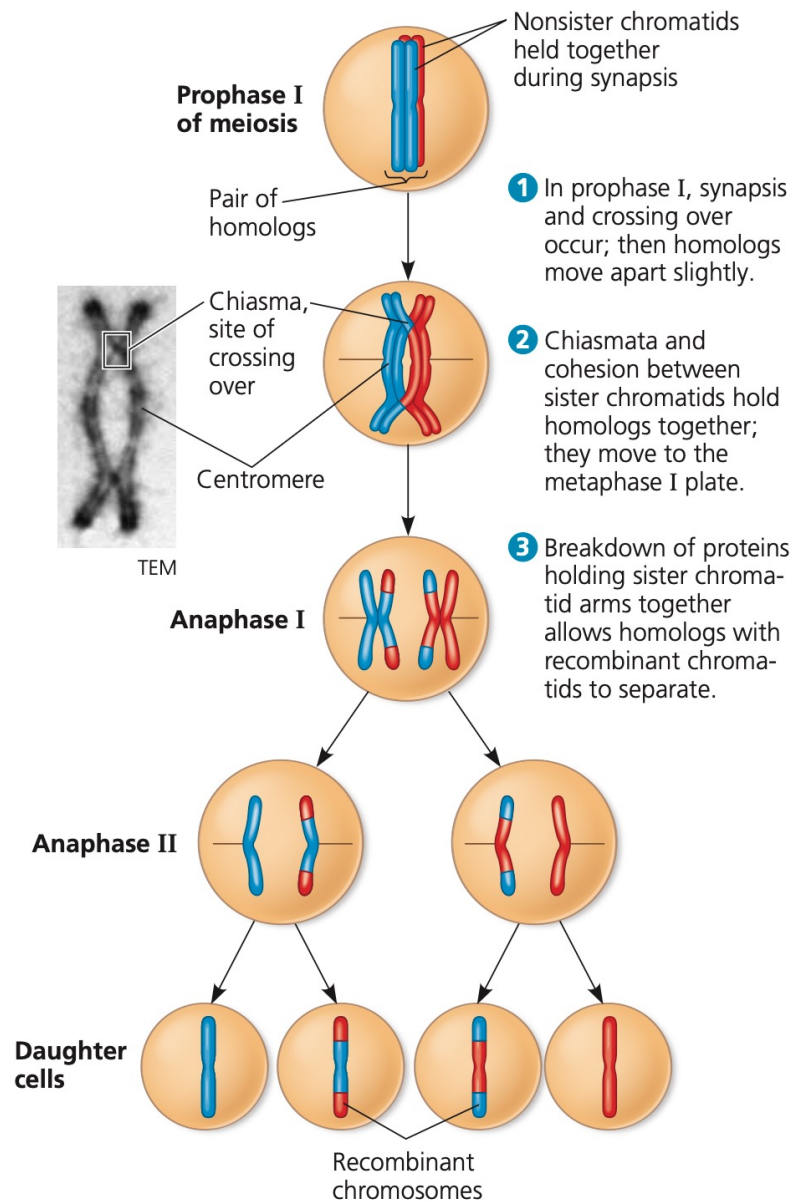
Individual Assortment

- Random orientation of pairs of homologous chromosomes at metaphase of meiosis I. At metaphase I, the homologous pairs, each consisting of one maternal and one paternal chromosome, are situated at the metaphase plate
- Each pair may orient with either its maternal or paternal homolog closer to a given pole (50% chance*)
- Hence, the first meiotic division results in each pair sorting its maternal and paternal homologs into daughter cells independently of every other pair

- Each daughter cell represents one outcome of all possible combinations of maternal and paternal chromosomes.
- In the case of humans ($n = 23$), the number of possible combinations of maternal and paternal chromosomes in the resulting gametes is 2^{23} , or about 8.4 million.
- This is still an underestimate as we have yet to take account of the other two factors that influence genetic variation.



▼ **Figure 13.12** The results of crossing over during meiosis.



Crossing Over

- During **prophase I**, homologous chromosomes pair up and undergo **synapsis**.
- Non-sister chromatids may exchange corresponding DNA segments.
- This process produces **recombinant chromosomes**, which contain both maternal and paternal DNA.
- Crossing over creates **new allele combinations** not found in either parent chromosome.

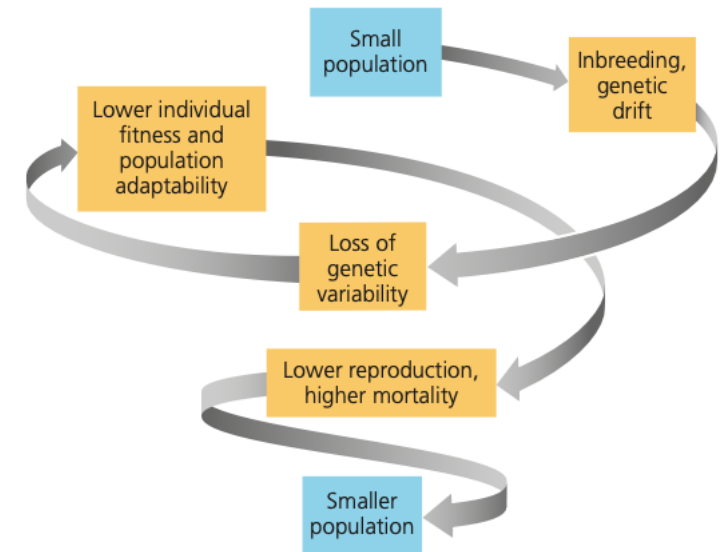
Random Fertilization

- Any sperm can potentially fertilize any egg.
- This means that gametes with many different genetic combinations can unite in numerous ways.
- Random fertilization increases variation even further after meiosis has already produced diverse gametes.

Why Genetic Variation Matters for Evolution

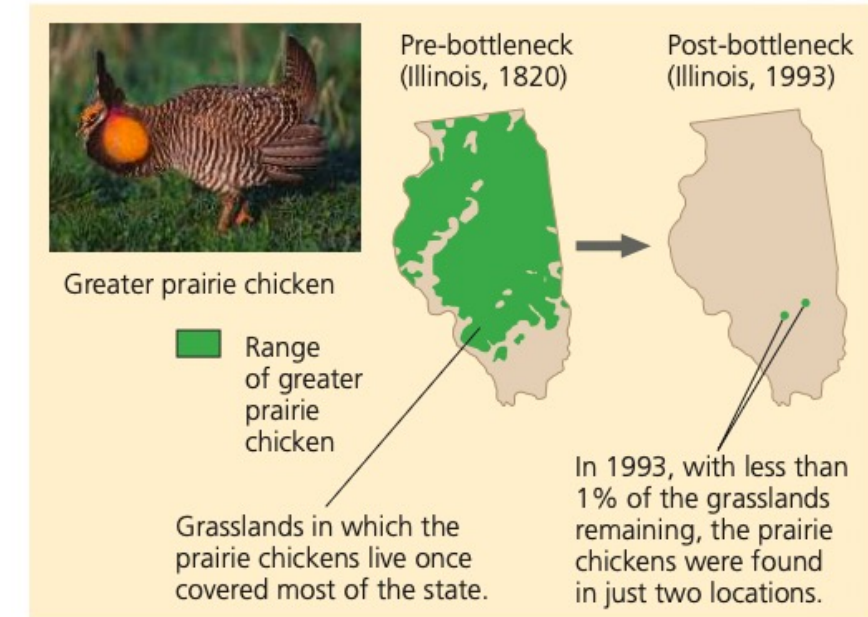
- Genetic variation is required for **natural selection** and other evolutionary mechanisms to act.
- Without heritable differences, allele frequencies cannot change meaningfully over time.
- Low genetic variation can reduce a population's ability to adapt to environmental change.
- Reduced variation can be associated with lowered fitness and reduced adaptability in small populations.

▼ Figure 56.11 Processes driving an extinction vortex.



Additional Pointers

- Genetic variation should be understood at both the **population** and **molecular** levels.
- Not all variation is equally important:
 - some is **neutral**,
 - some is **harmful**,
 - some may become advantageous under certain conditions.
- Sexual reproduction mainly reshuffles existing alleles, but long-term evolutionary novelty still depends on **mutation**.
- Thus, genetic variation is both:
 - the immediate basis of population-level evolution,
 - and the long-term substrate for the origin of new adaptations.



(a) The Illinois population of greater prairie chickens dropped from millions of birds in the 1800s to fewer than 50 birds in 1993.

Location	Population size	Number of alleles per locus	Percentage of eggs hatched
Illinois			
1930–1960s	1,000–25,000	5.2	93
1993	<50	3.7	<50
Kansas, 1998 (no bottleneck)	750,000	5.8	99
Nebraska, 1998 (no bottleneck)	75,000–200,000	5.8	96

(b) In the small Illinois population, genetic drift led to decreases in the number of alleles per locus and the percentage of eggs hatched.

Conclusion

- Genetic variation refers to differences in genes and DNA sequences among individuals.
- It may be measured through **heterozygosity** and **nucleotide variability**.
- New variation originates through **mutation** and related DNA changes.
- Sexual reproduction amplifies variation through:
 independent assortment,
 crossing over,
 and random fertilization.
- Genetic variation is essential because it provides the basis for **adaptation and evolution.**

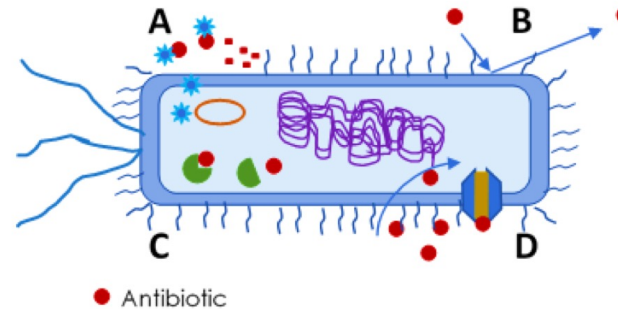
IBO past year questions

IBO 2021 Challenge II Theoretical Paper: Qn 36

Q36

Antibiotic resistance is a constant Public Health problem as more antibiotics are rendered ineffective by drug-resistant bacteria which have developed resistant mechanisms to conventional antibiotics.

Bacteria can resist antibiotics via different resistance mechanisms. In the **figure below**, identify the mechanisms **A – D** from the list 1 – 6: (4 points, 1 point each correct answer)



1. Modification of the antimicrobial drug target.
2. Production of hydrolytic enzymes, such as β -lactamases, that inactivate the surrounding antimicrobial drug.
3. Production of bacteriocins that destroy the antibiotic.
4. Reduction of the antimicrobial drug concentration in the intracellular environment by pumping them out before they can take effect.
5. Antibiotic degradation inside the cell
6. Natural resistance due to an unusually impermeable cell membrane or lack of the target recognized by the antibiotic

Answer Key

Q36

A.	2
B.	6
C.	1
D.	4